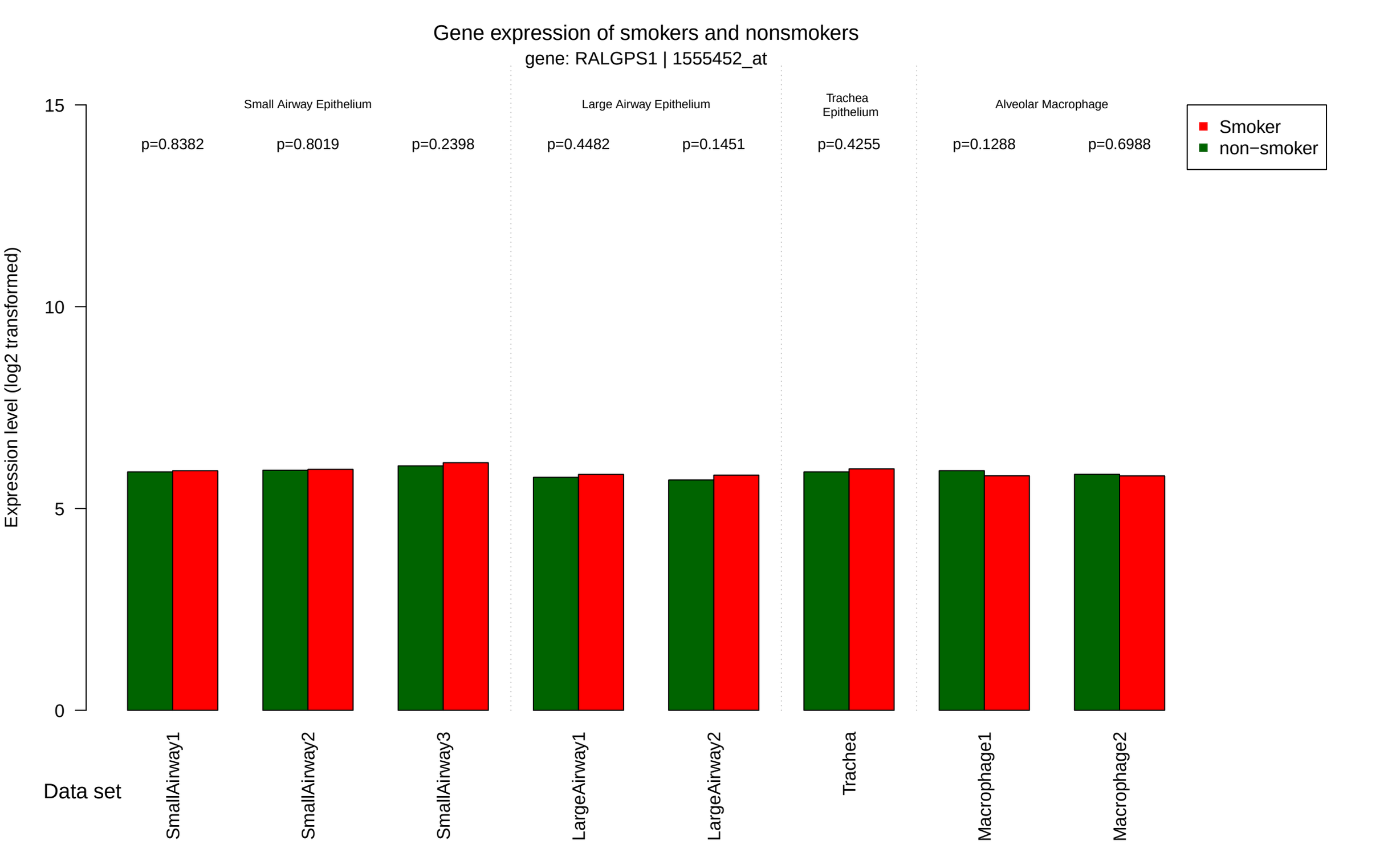
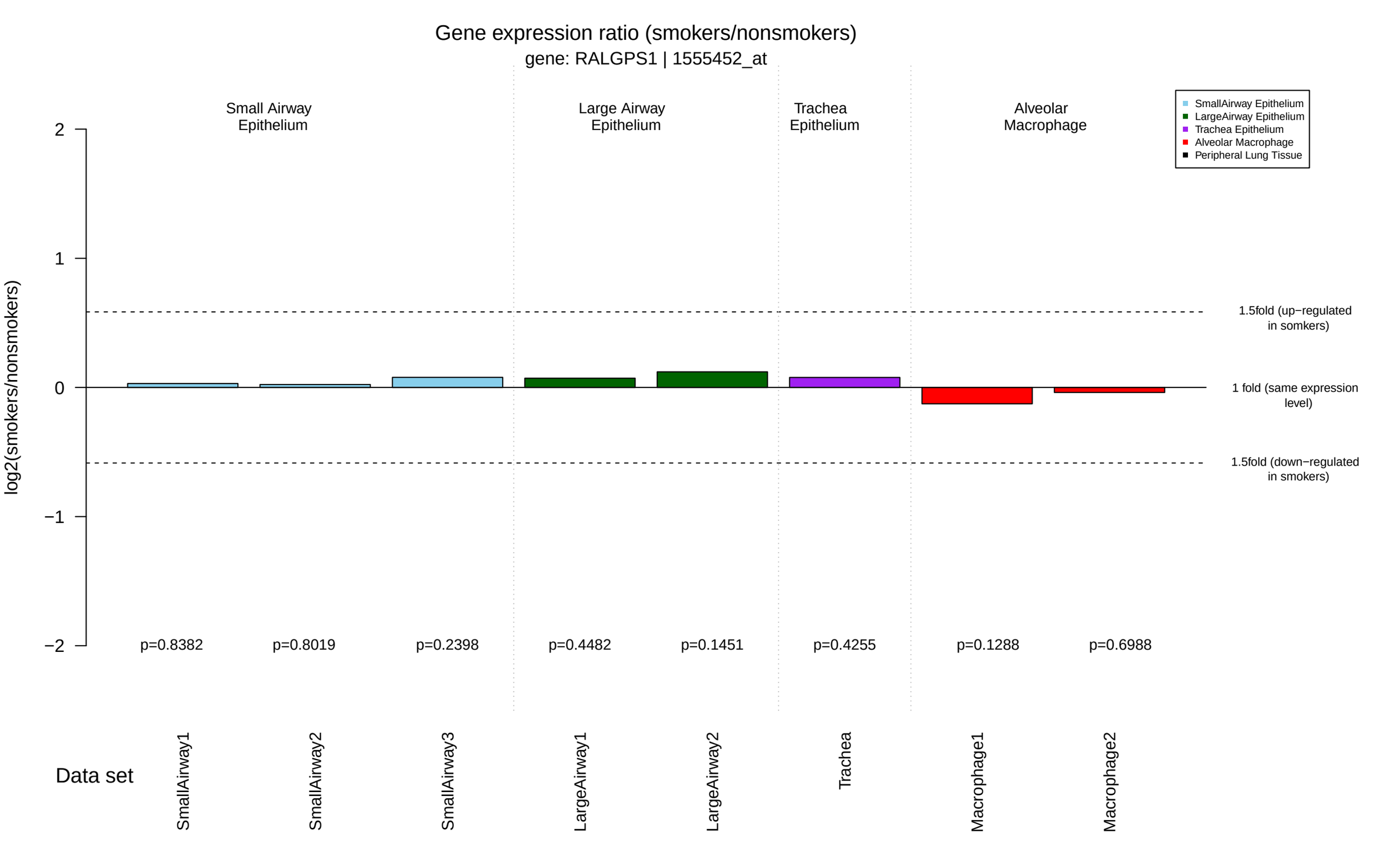
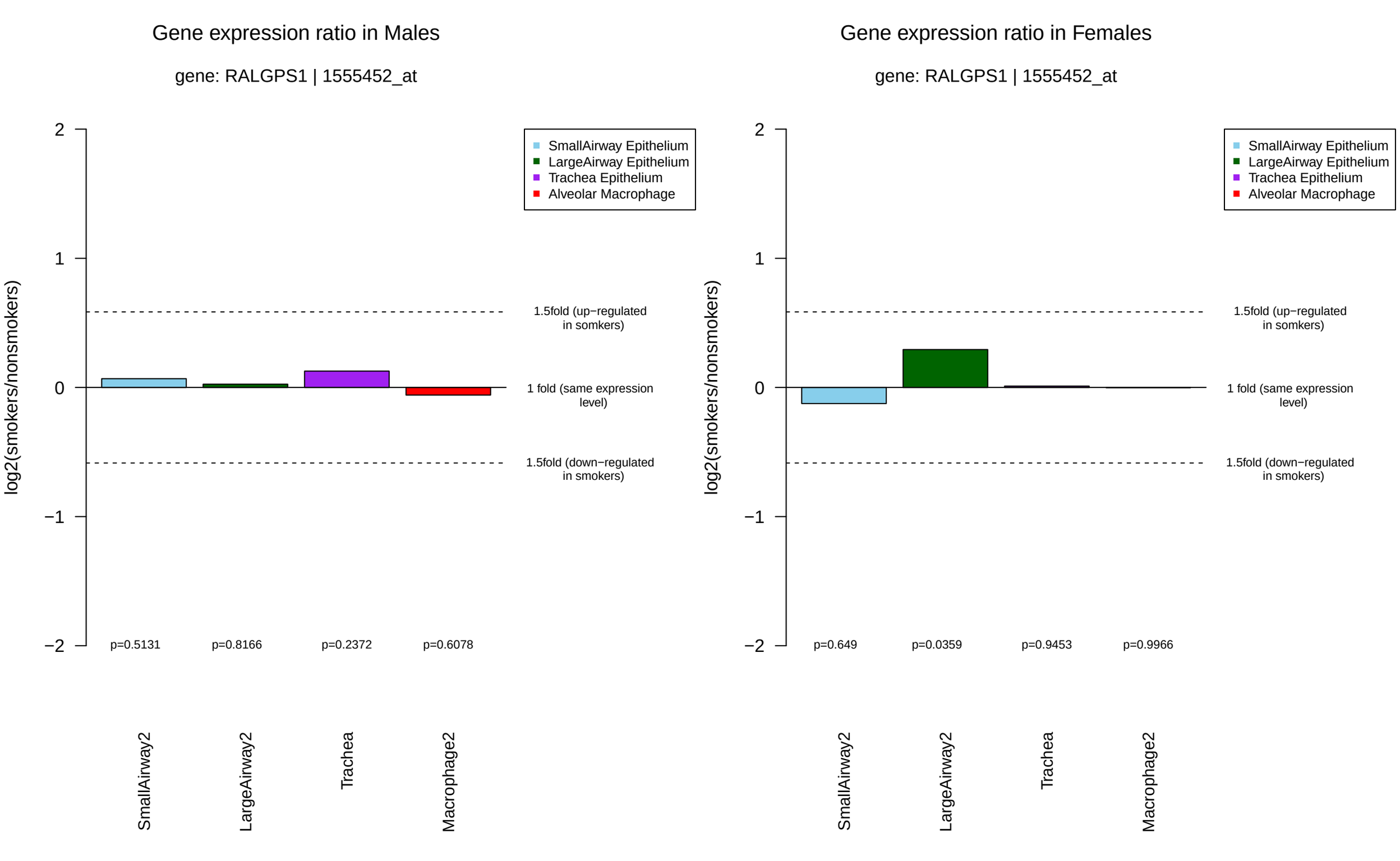
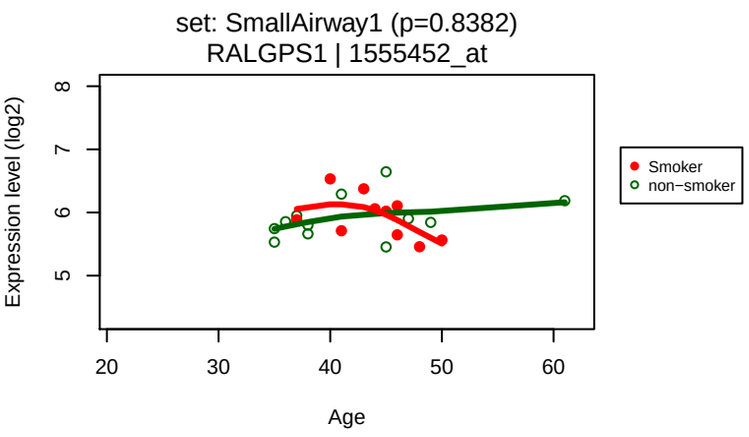
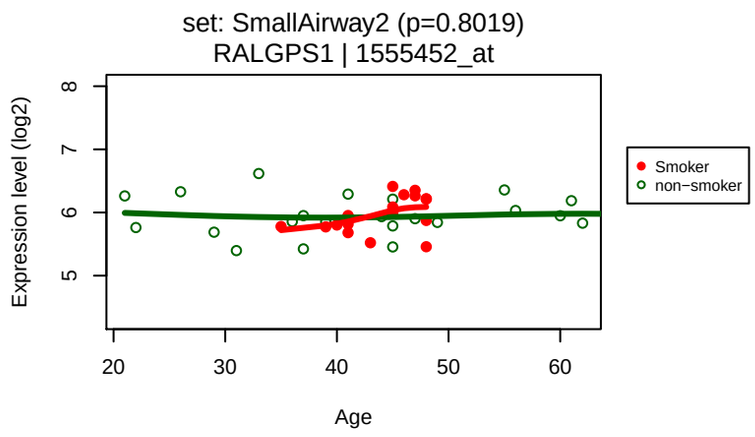
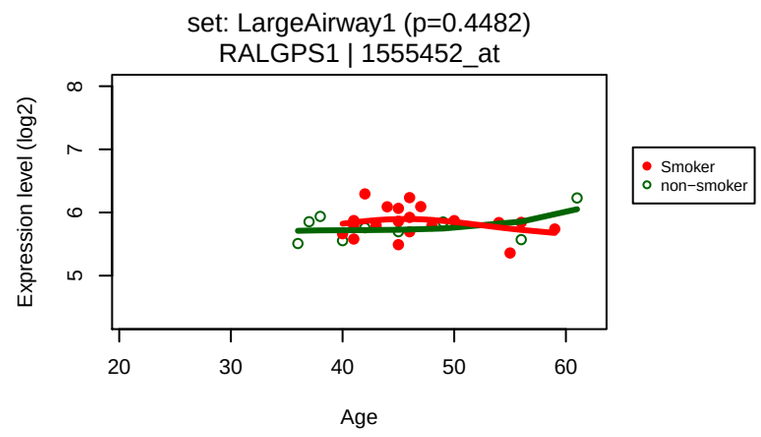

Gene expression of smokers and nonsmokers
gene: RALGPS1 | 1555452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8382
p=0.8019
p=0.2398
p=0.4482
p=0.1451
p=0.4255
p=0.1288
p=0.6988
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

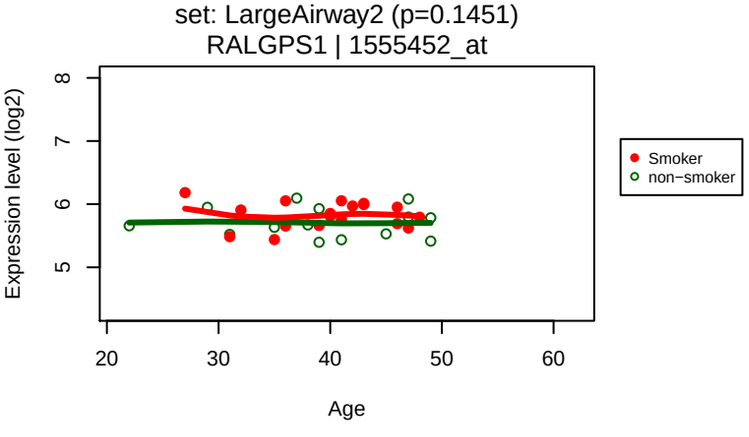

Gene expression ratio (smokers/nonsmokers)
gene: RALGPS1 | 1555452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8382
p=0.8019
p=0.2398
p=0.4482
p=0.1451
p=0.4255
p=0.1288
p=0.6988
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

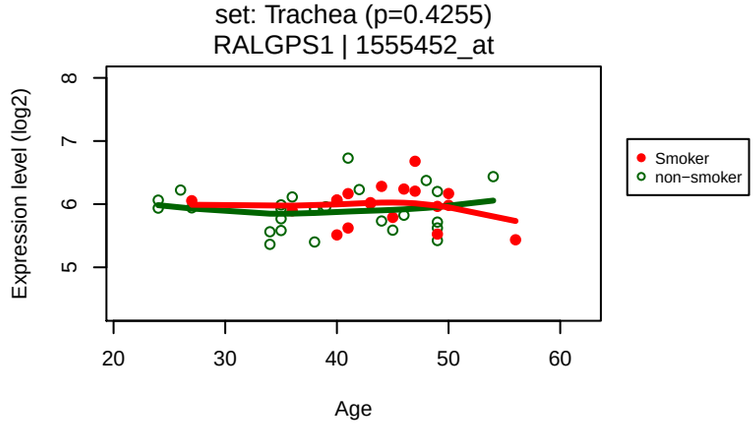

Gene expression ratio in Males
gene: RALGPS1 | 1555452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5131
p=0.8166
p=0.2372
p=0.6078
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGPS1 | 1555452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.649
p=0.0359
p=0.9453
p=0.9966
SmallAirway2
LargeAirway2
Trachea
Macrophage2

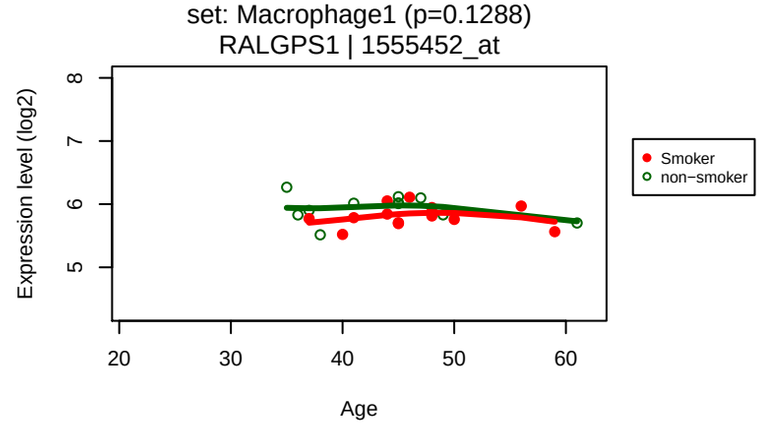

set: SmallAirway1 (p=0.8382)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker

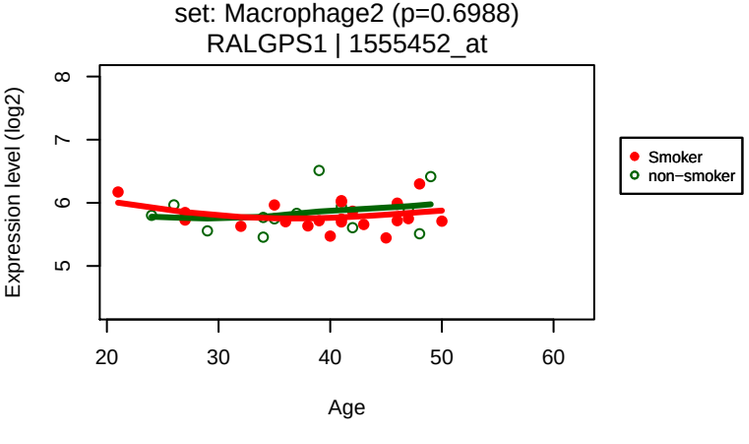

set: SmallAirway2 (p=0.8019)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker

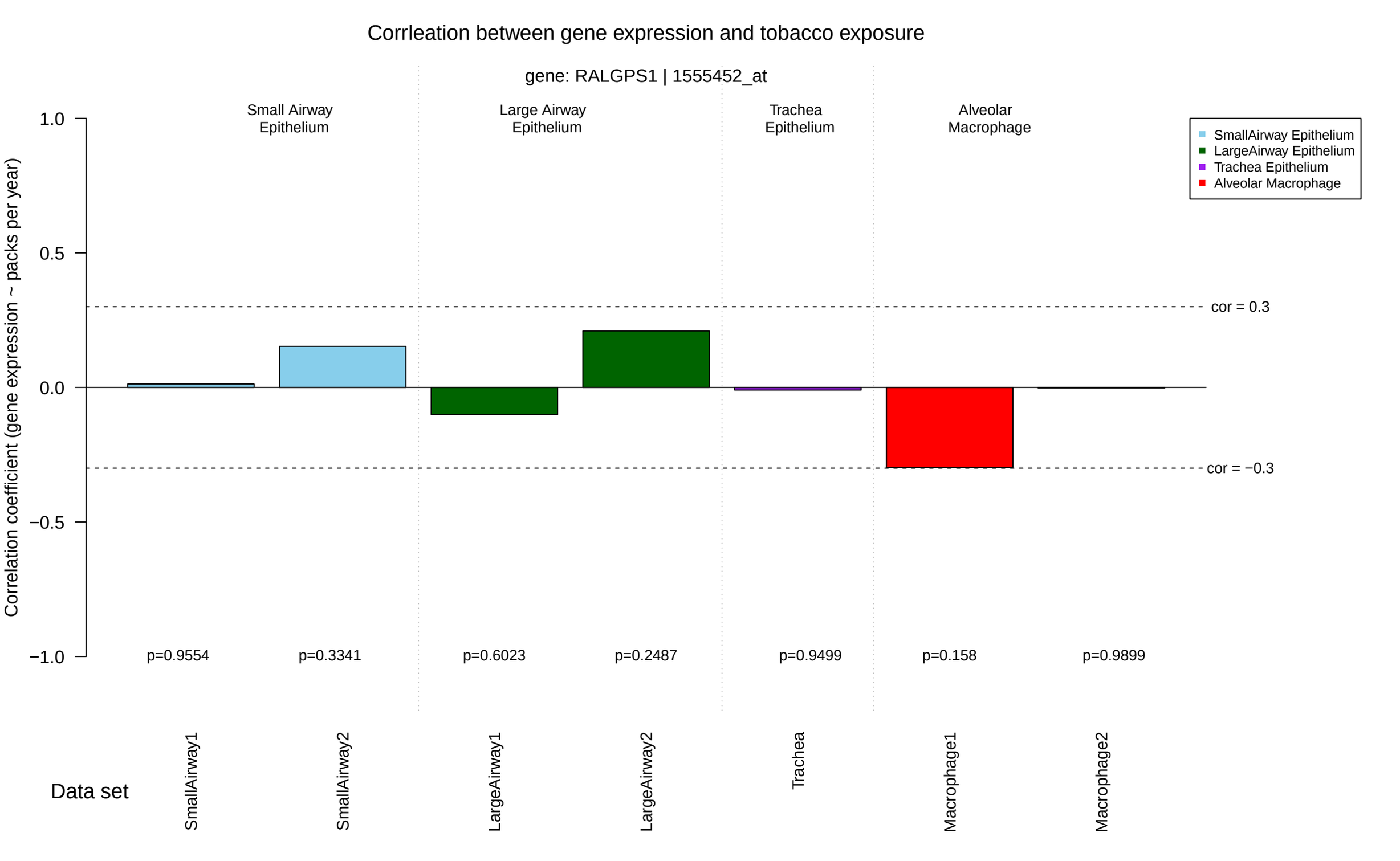

set: LargeAirway1 (p=0.4482)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1451)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4255)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1288)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6988)
RALGPS1 | 1555452_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGPS1 | 1555452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9554
p=0.3341
p=0.6023
p=0.2487
p=0.9499
p=0.158
p=0.9899
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2